



Pragati
Engineering
College
(Autonomous)

Library Management System

Pragati Engineering College

Presented by:

G. Tejaswini
25A31A4307

N. Krishnaveni
25A31A4322

M. Pavan satish
25A31A4352

P. Harika
25A31A4323

K. Krupa varma
25A31A4347

What is an LMS?

A Library Management System (LMS) is an enterprise resource planning software specifically designed for libraries.

It automates the core functions of a library, including acquisition, cataloging, circulation, and reporting.

Goal: To streamline operations, enhance user experience, and ensure efficient tracking of resources.



System Process Flowchart

Logical Data Flow

The flowchart illustrates the logical flow of data within the system.

- **User Login:** Authentication gateway.
- **Search:** Querying the book database.
- **Transaction:** Issue or Return processing.
- **Database Update:** Real-time inventory adjustment.

 Library Management System Flowchart



Core System Modules



Admin

Manages entire system, adds librarians, generates high-level reports, and maintains system settings.



Librarian

Handles book issues/returns, manages student records, calculates fines, and updates book inventory.



Student

Searches for books, views borrowed history, checks due dates, and reserves books online.

Manual vs. Automated System

Feature	Manual System	Automated LMS
Speed	Slow and time-consuming	Instant retrieval of data
Accuracy	Prone to human errors	High accuracy and consistency
Data Security	Physical records can be lost/damaged	Secure digital backups available
Searchability	Difficult to find specific records	Advanced filtering and search

Book Circulation Workflow



Search

Student locates book in
OPAC



Request

Book issue request to
librarian



Issue

System updates inventory
& user account



Return

Book check-in & fine
calculation

Database Schema: 'Books' Table

Field Name	Data Type	Constraint	Description
Book_ID	VARCHAR(10)	PRIMARY KEY	Unique identifier for book
Title	VARCHAR(100)	NOT NULL	Name of the book
Author	VARCHAR(50)	-	Name of the author
Status	VARCHAR(15)	CHECK	'Available' or 'Issued'

Key Benefits

-  **Time Efficiency:** Reduces administrative time by over 70%.
-  **Data Integrity:** Ensures records are synchronized.
-  **Cost Effective:** Reduces paperwork overhead.
-  **Better Analytics:** Insights into borrowing trends.

Technologies Used



Frontend

HTML5, CSS3, JavaScript



Backend

C Language (Logic), File Handling



Data

Structs & Arrays (In-Memory)

Future Enhancements

Mobile Application: Developing an Android/iOS app for students to manage renewals on the go.

AI Integration: Implementing a recommendation engine to suggest books based on reading history.

RFID Integration: Moving towards fully automated self-service kiosks for check-out.



System Implementation (C Language)

```
#include
#include

struct Book {
    char name[50];
    int quantity;
};

int main() {
    struct Book library[100];
    int n, i, totalQuantity = 0;

    printf("Enter number of different books: ");
    scanf("%d", &n);

    for (i = 0; i < n; i++) {
        printf("\nEnter name of book %d: ", i + 1);
        scanf("%s", library[i].name);
        printf("Enter quantity of book %d: ", i + 1);
        scanf("%d", &library[i].quantity);
        totalQuantity += library[i].quantity;
    }
```


Questions?

Thank you for your attention.

Pragati Engineering College

www.pragati.ac.in

Image Sources



<https://admission.pragati.ac.in/wp-content/uploads/2025/05/Pragati-Logo-JPEG-01.jpg>

Source: admission.pragati.ac.in



https://png.pngtree.com/thumb_back/fh260/background/20251128/pngtree-modern-library-interior-with-students-studying-at-long-tables-and-bookshelves-image_20639507.webp

Source: [pngtree.com](https://png.pngtree.com)



<https://svg.template.createely.com/jbgiyigl>

Source: [createely.com](https://svg.template.createely.com)



<https://media.istockphoto.com/id/1218534419/vector/online-library-and-media-books-archive-concept-tiny-people-characters-at-huge-computer.jpg?s=612x612&w=0&k=20&c=bM5IIVASSEVnoaFC3UnlyMyO4McGjUYAbajflWUPPtY=>

Source: www.istockphoto.com